The head-start cap: an empty sky under a shared-onset lineage clock

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Abstract

We are close to being able to expand beyond our own planet, and we see nobody else doing it. The hard-steps model explains the silence by making civilisations rare: a chain of improbable transitions on each planet, with earlier expanders closing off territory for everyone else. Mills et al. (2025) have argued that Earth's own record gives poor support for improbable steps, since the candidate transitions may simply have waited for conditions to change. Take that seriously and a problem appears. If lineages usually get there in the end, the older ones had a large head start, and some of those should now be visible. We work out how large that head start is allowed to be. Using flat-space geometry and perfect detection of expanded galaxies, we find that a clock tied to each planet's birth needs the mean number of loud lineages per galaxy, $\Lambda = qN$, held below $\sim 2.1 \times 10^{-8}$. A clock that starts from a shared cosmic event and carries progress between planets faces a looser constraint. With a single shared emergence date and Λ large enough that most galaxies contain a loud lineage, the permitted lead is about 14 million years. Spreading the start times over a window forces our own start toward the front of that window. Independent, normally distributed lineage durations tighten the constraint further: with a standard deviation of 1% of 13.6 Gyr and $q = 1$, about 6×10^{-12} of zones have emerged by today, and we sit in the first flush of arrivals inside our own galaxy as well as outside it.

1. The observation

Start with something we can see. Humanity is one technological civilisation, arriving at cosmic age $t_0 = 13.8$ Gyr, in a sky with no sign of a civilisation that has spread across its galaxy. Any account of where civilisations come from has to produce that sky, seen from here, now.

The hard-steps account produces it through rarity. A lineage must pass n transitions whose typical waiting times exceed the life of its planet, so the chance of emerging within time t grows as t^n and almost all planets never get there (Carter 1983; Watson 2008; Snyder-Beattie et al. 2021). Hanson et al. (2021) add expansion: the rare lineages that emerge and spread visibly ("grabby" civilisations) occupy volume where later ones can no longer appear independently, so anyone who finds themselves in an empty sky arrived before the local deadline.

Mills et al. (2025) pull in a different direction. Working through the usual candidate steps, they argue that happening once on Earth is weak evidence of improbability. The first lineage to cross a threshold tends to close the niche behind it, and several thresholds coincide with planetary changes (rising oxygen is the standard example) that arrived late for reasons having nothing to do with biological luck. On their reading, Earth's record looks like a sequence of doors opening.

Earth's record, though, is one planet. Suppose the doors do open in sequence and most suitable lineages walk through them eventually. Older planets had more time: in the Galactic model

of Lineweaver, Fenner, and Gibson (2004), about 75% of stars in the Galactic habitable zone are older than the Sun. Hanson posed the resulting problem for easy spread followed by easy evolution in 2010 (Hanson 2010) and returned to it after Mills et al. (Hanson 2024). It is the question this note takes up: how much sooner can older lineages have emerged, given that we see none of them?

The clock we test runs as follows. Progress accumulates in lineages that can move between environments, so a planet's birth starts nothing. All lineages draw on progress from a shared cosmic starting point (our candidate is the first appearance of carbon chemistry) and take a characteristic total time from there. Pieces of this picture are old: Hanson (1998) allowed one difficult beginning followed by steady advance; galaxy-wide synchronisation has its own literature (Annis 1999; Ćirković and Vukotić 2008); Sharov and Gordon (2013) argued for a lineage older than Earth. Ponfilenok (2026) works a related model inside one galaxy, with synchrony arising from many small steps; here the bound comes from expansion fronts between galaxies.

We offer no odds on the process clock against hard steps. The calculation assumes the process clock and finds what it costs. The accounts differ in where they put the improbability: inside the transitions, or in the timing.

2. Setup

Count suitable spacetime in Earth-equivalent units. A **zone** here is a run of conditions in which evolution can proceed and retain its gains — liquid, an energy gradient and churn, on an icy body as readily as on a planet. Suitability is relative to the process's current state: the instantaneous arena a lineage needs grows as it complexifies, so small wet bodies qualify early and only larger arenas, eventually planets, qualify late. Let N be the effective number of retained-progress trajectories still in play in a galaxy today. We take $N = 10^{10}$ as the central value, normalised for scale as 10^{11} stars times 0.1 effective trajectories per star, and run the calculation across 10^8 to 10^{11} where it matters. That normalisation is a Fermi scale, not a count of Earth-like planets. The counted trajectories $N(t)$ grow from near zero at cosmic age 0.2 Gyr, tracking the growth of effective Goldilocks capacity as enrichment spreads. They are not planets being born. A counted trajectory is assumed to keep finding an arena large enough as the requirement grows; dropping out when a site is outgrown is left out of this calculation.

Count one lineage per effective trajectory. Two clocks (Figure 1):

- *Planet clock.* For this comparison only, identify each trajectory with one planet. A lineage on a planet born at time b becomes able to begin loud expansion — crosses the loud threshold — at $b+T$. We call that crossing **emergence**; it names an operational endpoint, not a biological stage.
- *Inherited clock.* Progress runs from a shared start s. A trajectory already in play when the running total reaches T crosses at $s+T$. A trajectory entering the count at $b < s+T$ joins the frontier wherever it has got to and also crosses at $s+T$. One entering at $b > s+T$ is credited with the crossed state on entry, giving crossing time $\max(b,\, s+T)$. Read that formula as bookkeeping for a shared frontier. An arena-resolved version would follow a trajectory between sites as the arena it needs grows; the one clock compresses the whole history into T.

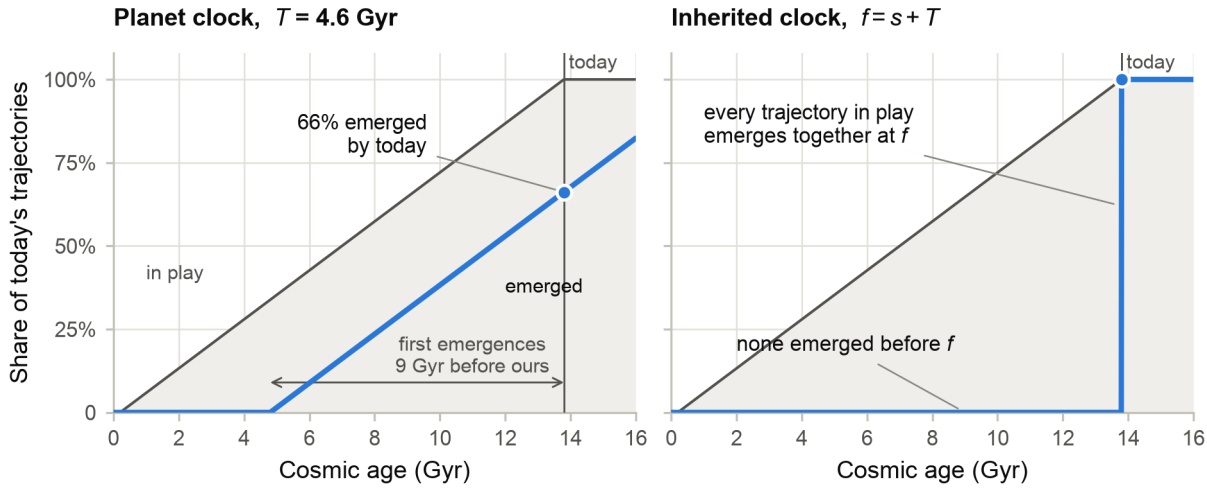


Figure 1: Two clocks, same trajectories. Grey: share of today's effective trajectories in play at each cosmic age (a linear ramp from 0.2 Gyr). Blue: share that has crossed the loud threshold. Left, the planet clock at $T = 4.6$ Gyr: crossing trails a planet's birth by T, so two thirds of trajectories have crossed by today, the first 9 Gyr before us. Right, the inherited clock: nothing crosses before the shared date $f = s + T$, when every trajectory in play by then crosses at once in this one-clock bookkeeping. Drawn with $f = t_0$, so $\delta = 0$.

A lineage that has crossed turns loud (spreads through its galaxy and keeps spreading) with probability q. It becomes visible to us later still, after the front has crossed its galaxy and the light has travelled. The front moves at $v = 0.5c$. We mostly need one combination:

$$\Lambda = qN,$$

the mean number of lineages per galaxy that cross the loud threshold and turn loud, if every counted trajectory crosses. Call it the loud yield. With m crossed lineages in a galaxy, the chance that at least one is loud is $1-(1-q)^m$, close to $1-e^{-qm}$ whenever q is small. Λ governs most of what follows; q and N stay separate in the appendix and where the difference matters.

The geometry follows Olson (2018) and Olson and Ord (2021), simplified. Space is flat. Galaxies are scattered independently at an average density of 0.01 per cubic megaparsec: pick any volume and the galaxy count averages $0.01\times$ volume, with Poisson scatter around it. Two ways for a predecessor to have shown itself:

- *We could see it.* Once a front has crossed its home galaxy (30 kpc across, about 0.2 Myr at $0.5c$), the galaxy counts as visible, and its light needs r/c to reach us from distance r.
- *It could be here.* The front itself reaches us after r/v.

For each condition separately, count the galaxies expected to satisfy it and call the count μ. Independent galaxies give an empty-sky probability of $e^{-\mu}$. Our working line is $\mu = -\log 0.05 \approx 3$: beyond it, an empty sky has under a 5% chance *inside this model*. The inputs are assumed, several guessed to an order of magnitude, so each bound below is an edge for those inputs. Giving a bound a probability in the real world would take priors over the inputs and a rule for sampling observers.

Detection here is perfect: every crossed galaxy registers. Existing waste-heat searches look for galaxies re-radiating a large share of their starlight in the mid-infrared, in samples of limited size and depth (Wright et al. 2014; Griffith et al. 2015). Our μ counts loud galaxies present in

the volume. Turning it into a survey probability would need the intercepted starlight fraction, distances, dust and catalogue completeness. Appendix A collects the equations.

3. If every planet starts its own clock

This section imposes the opposite rule from the one we favour, to see what it costs. Read each counted trajectory as one planet, starting from nothing at that planet's birth and needing the same time T. Earth caps T at 4.6 Gyr for any history run entirely on one planet, and 4.6 Gyr is the choice friendliest to this branch, since a shorter T moves every other planet's emergence earlier as well.

On the same ramp, counted planets have been forming since 0.2 Gyr — earlier than rocky planets were plausibly common, which is again generous to this branch. At $T = 4.6$ Gyr every planet born before cosmic age 9.2 Gyr has emerged: two thirds of today's stock, with the first emergences 9 Gyr before ours. That count keeps every planet suitable until emergence. There is no destruction, no stellar-lifetime cut and no closing habitable window. A cut would remove some old planets before they could emerge and soften the ceiling below, while leaving a spread of billions of years among the survivors. Holding these emergences to an empty visible sky requires $\Lambda \lesssim 2.1 \times 10^{-8}$. In words: fewer than one galaxy in fifty million may produce a loud lineage, ever, out of its 10^{10} counted trajectories ($q \lesssim 2.1 \times 10^{-18}$ at the central N). Counting only planets old enough to have emerged by today, about one galaxy in 73 million hosts an expander now. The arrival condition gives a similar ceiling.

Trust the shape of this result more than its digits (Figure 2). It integrates over gigaparsecs and most of cosmic history, where flat space and a fixed galaxy density are crude. The shape says: planet clocks plus common emergence leave only one place to hide, inside q. Almost every emerged lineage has to stay quiet, die young, or expand in a way nobody can see. Something has to filter them.

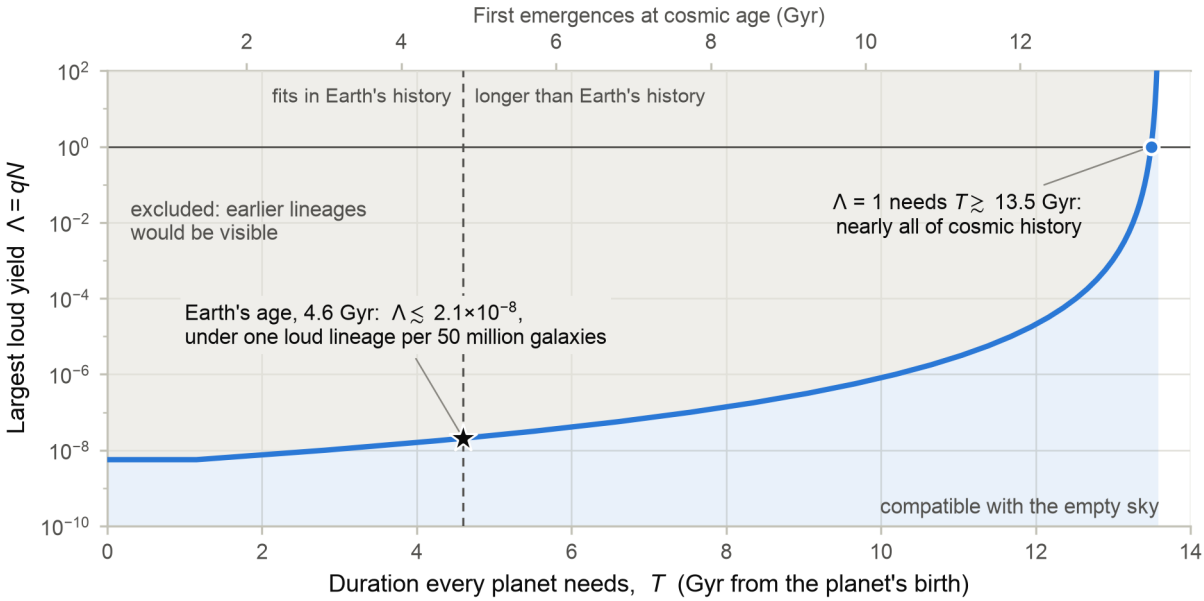


Figure 2: Planet clocks. The largest loud yield $\Lambda = qN$ compatible with an empty sky ($P(\text{empty}) \geq 5\%$, visibility at $v = 0.5c$), against the duration T every planet needs; the top axis gives the cosmic age of the first emergences, $0.2\,\text{Gyr} + T$. Grey: excluded. At Earth's age, 4.6 Gyr, the ceiling is $\Lambda \approx 2.1 \times 10^{-8}$. Yields of order unity need durations approaching the age of the universe.

4. A shared starting gun

Now let progress carry between planets, running from the shared start s. Planet age stops mattering: what matters is the date $f = s + T$ at which the process first emerges anywhere, and how far that date sits ahead of ours. Write the lead as $\delta = t_0 - f$.

Take the extreme case first: every lineage shares the same start and duration, so one frontier date f describes all of them. This case cannot describe our own position (under it we emerge at f too, and $\delta = 0$). It is the limit the spread-out versions approach, and it shows the geometry cleanly.

For $\Lambda \gtrsim 10$ the visibility condition gives $\delta \lesssim 13.7$ Myr (Figure 3). Arrival allows 27 Myr; tightening the line to $\mu = 1$ gives 9.6 Myr. At $\Lambda = 1$ and 0.01 the leads are 16 and 63 Myr. The plateau is a volume statement: a lead δ exposes loud galaxies out to about $c\delta$, and at 13.7 Myr that sphere reaches ~ 4 Mpc and holds about three galaxies. Once those galaxies are loud, more lineages per galaxy change nothing (a galaxy cannot be louder than loud; Figure 4). Cosmic expansion is negligible over 4 Mpc, which is why this number is firmer than Section 3's.

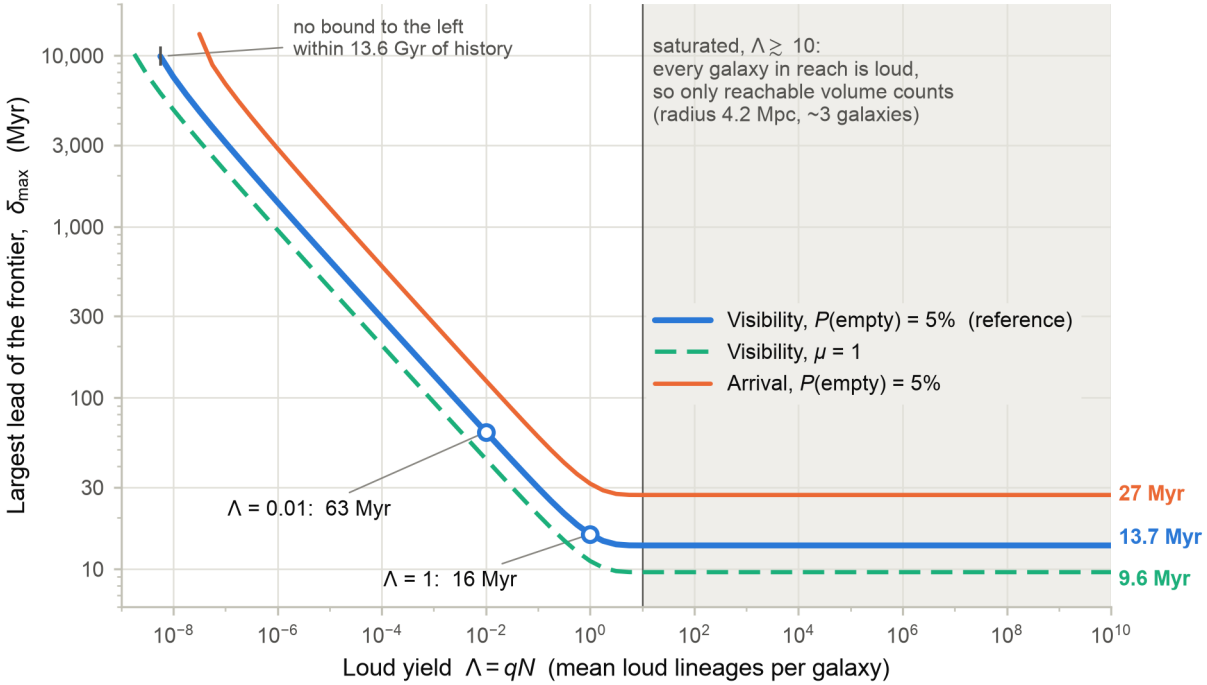


Figure 3: The head-start cap. Largest lead δ of a shared emergence frontier over our own emergence, against loud yield Λ, for three sky conditions. In the grey band ($\Lambda \gtrsim 10$) every galaxy in reach is loud and the cap is set by volume alone: 13.7 Myr for visibility at $P(\text{empty}) = 5\%$, 9.6 Myr at $\mu = 1$, 27 Myr for physical arrival. Left of the tick, no lead within the 13.6 Gyr history is excluded.

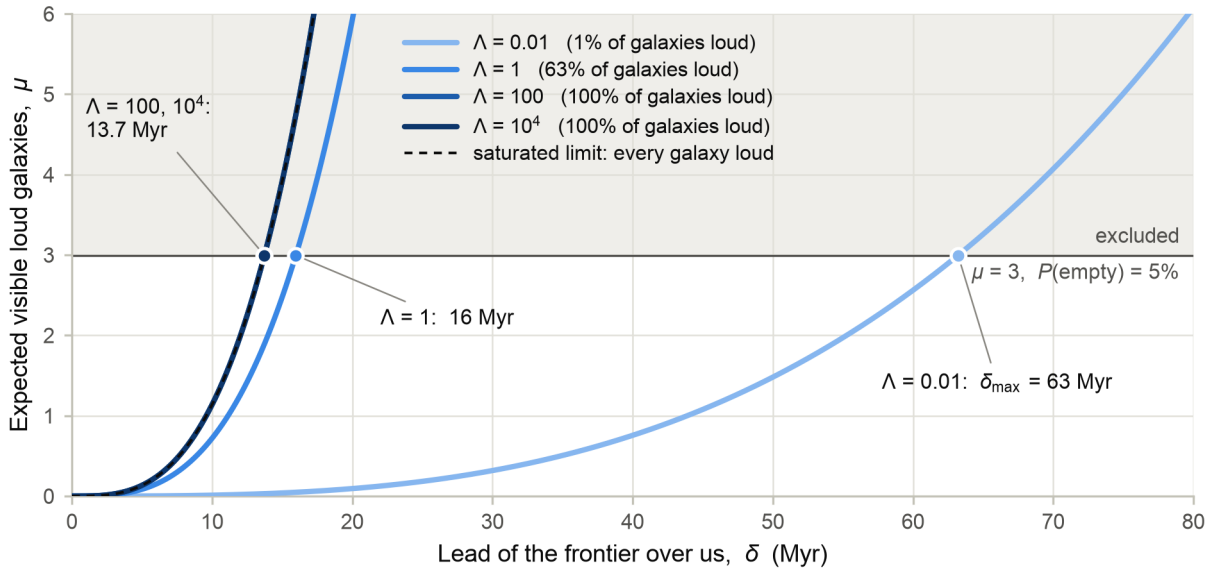


Figure 4: Why the cap flattens. Expected visible loud galaxies μ against lead δ at four loud yields, darker for larger Λ. Dashed: the saturated limit with every galaxy loud, $\mu = \frac{4\pi}{3} n_g [c(\delta - D/v)]^3$. At $\Lambda = 100$ and 10^4 the model curves lie on it, so extra lineages change nothing. Each $\delta_{\max}$ is where a curve crosses $\mu = 3$; grey is excluded.

Real starts will scatter. Let the start of each galaxy's clock fall somewhere in a window of width W, all durations still equal, expansion saturated (Λ large). Emergence order then copies starting order. We emerged today; the galaxy that started first emerged δ before us, where δ is exactly our own distance into the window. The sky caps δ, and the cap translates into a position inside the window (Figure 5):

Window W	Largest δ	Our start must lie within
100 Myr	32 Myr	first 32%
200 Myr	38 Myr	first 19%
500 Myr	47 Myr	first 9.5%
5 Gyr	84 Myr	first 1.7%

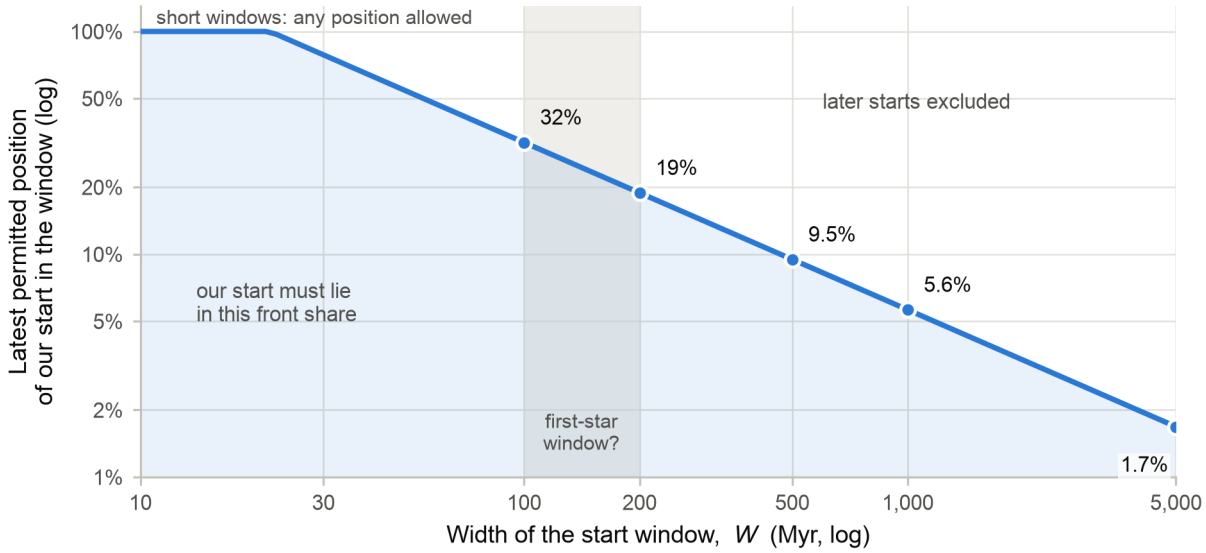


Figure 5: Spreading the start. Latest permitted position of our start inside a uniform start window of width W, with saturated yield and visibility at $P(\text{empty}) = 5\%$. Below about 20 Myr any position is allowed; past that, widening the window shrinks the permitted share almost as fast. Grey band: an illustrative 100–200 Myr window for first-star enrichment. Each point is an edge in the model, not a probability.

Read the right column as an edge, in the sense of Section 2: start later than it and an empty sky drops below a 5% chance in the model. It is not a probability that we really started inside the edge; that would need priors and an observer rule. Widen the window and the bill arrives as a position nearer its front. Fix our start at the middle of a 5 Gyr window instead, and the bill arrives as $q \lesssim 1.4 \times 10^{-16}$, back in filter territory.

What could fire starting guns across many galaxies within a few hundred Myr? It must reach everywhere and switch on conditions for accumulation. An earlier universal switch exists on paper: for roughly 7 Myr, when the universe was 10–17 Myr old, the cosmic microwave background alone kept surfaces between freezing and boiling (Loeb 2014). That Goldilocks interval was probably wasted. Temperature was right before the ingredients were: in the standard picture almost no star-forming haloes, heavy elements or rocky planets yet existed to use the warmth. The first stars and their carbon are the first gun with material to work on (Bromm 2013; Stacy and Bromm 2014). The sky calculation cannot select it: sliding the start date and adjusting T leaves all emergence dates unchanged. Halo assembly and mixing will widen whatever window the first stars set, so the 100–200 Myr rows are a case to test against enrichment histories.

Picture the two clocks running together. Early stages (chemistry, protocells) ask little of their surroundings: liquid water and an energy gradient, available on icy moons and small bodies as well as planets, and far more numerous than Earth-like zones. Progress there is set by the frontier, the best stage reached anywhere within reach, not by the age of any one body. Later stages need bountiful Goldilocks environments like Earth's surface. A planet that opened early gains little from waiting: until the frontier reaches a stage that can use it, an older Earth is one more site for early chemistry. Earth's LUCA, around 4.2 Gyr ago (Moody et al. 2024) and shortly after Earth formed, is on this reading close to a coincidence. It dates when frontier material met a suitable planet, not an origin. For timing, what matters is when the frontier can first use Earth-like zones, and how reliably the later stages then run. That meeting still requires transfer and establishment, whose rates may be low (Adams and Napier 2022); rock exchange inside a birth cluster could correlate sibling systems (Belbruno et al. 2012), without setting the frontier itself.

From a start at cosmic age 0.2 Gyr, our elapsed duration is 13.6 Gyr: the start choice read back to us. The next section moves the *typical* duration off that value.

5. Variation in duration

So far every lineage in a galaxy has shared one start and one duration. Split the two:

$$C_{i,g} = S_g + D_i,$$

with S_g a start shared across galaxy g and D_i the duration of lineage i. Section 4 priced scatter in S_g. To isolate duration scatter below, set $S_g = S$: one cosmic starting gun for every galaxy. A galaxy goes loud through its earliest loud lineage. How much earlier that happens depends on how independently its lineages develop. Shared ancestry and exchange of material can correlate their progress. The estimates below take durations as independent between zones, so each zone is a separate chance to emerge early. That is the severe end for precedence and the case our preferred reading must survive.

One choice drives everything in this section: the left tail of the duration distribution. The fitted percentile sits near 10^{-12} — about 6.8σ out on a normal curve — where no physical argument supplied here constrains the shape. Distributions that look alike near their centre can differ by orders of magnitude out there. A normal tail thins forever and always permits a few much faster lineages. A duration law with a hard physical floor $D_{\min}$ permits none before $S + D_{\min}$; if that floor date falls close to today and probability accumulates slowly just above it, as it would for many sequential stages, emergences can bunch near the frontier even with independent durations. A floor alone does not create that bunching (the density just above it matters), but it removes the long early tail that drives the extreme ranks below. Read the normal rows as arithmetic for one tail shape, not as a measurement of duration spread in general.

Give each lineage an independent duration, normally distributed with spread σ_T. Hair (2011) spread emergence times normally for up to 10^4 civilisations and stressed the first one's long head start, placing the centre of his distribution near today. Invert it: the first emergences arrive around now, so where does the centre sit? The fastest of 10^{10} draws lands about $6.4\sigma_T$ ahead of the mean. Holding the mean at or before today and using the 14 Myr plateau as a rough allowance for the lead gives $\sigma_T \lesssim 2$ Myr: durations equal to one part in 10^4. Local chemistry, waits for planetary conditions and sterilising accidents (Gowanlock, Patton, and McConnell 2011) will not hold that line over 13.6 Gyr.

Take a spread of hundreds of Myr and solve for the mean that keeps the sky empty, building each galaxy's loud probability from its 10^{10} draws. Then check home. The setup so far counted other galaxies; inside the Milky Way the expected loud lineages ahead of us are $\Lambda F(t_0)$, where F is our emergence percentile, and our chance of being first loud at home is $e^{-\Lambda F(t_0)}$. (Fronts cross the galaxy in under a million years, so travel time barely moves these rows.) At $q = 1$:

Duration spread σ_T	Share of zones emerged by today, $F(t_0)$	Expected loud lineages ahead of us in the Milky Way
30 Myr	5.9×10^{-10}	5.9
136 Myr (1% of 13.6 Gyr)	6.2×10^{-12}	0.06
408 Myr (3%)	2.7×10^{-13}	0.003

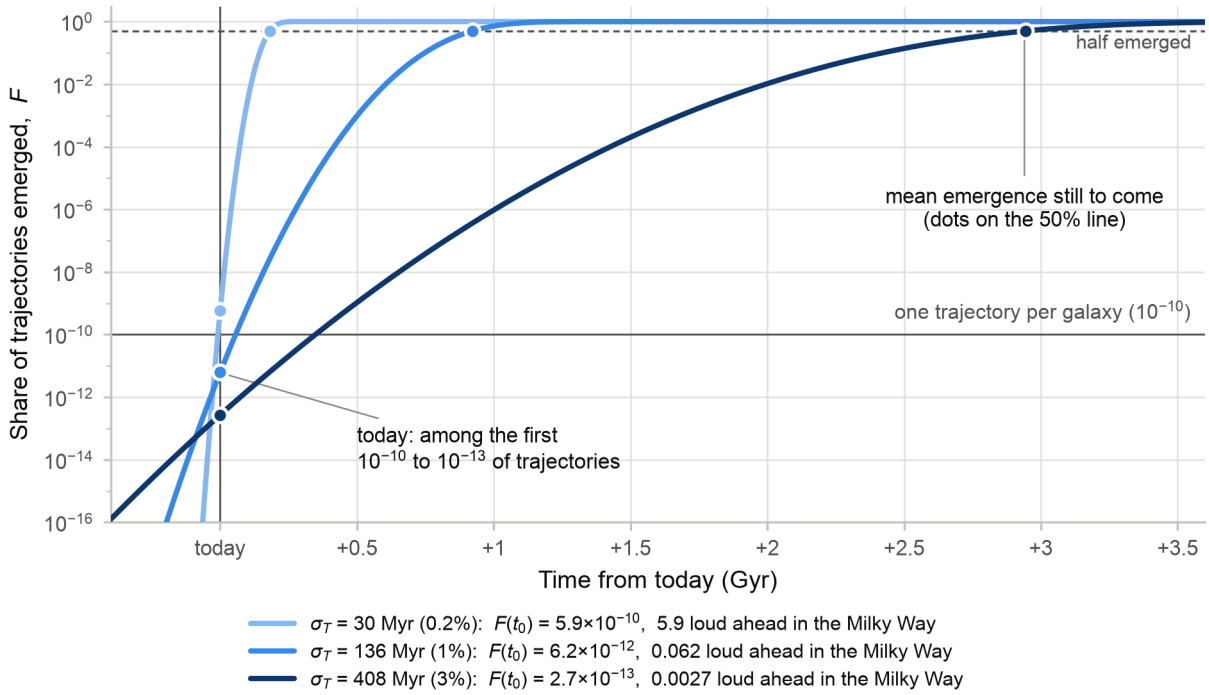


Figure 6: The first flush. Share of zones emerged, F, against time from today, for three spreads of independent normal durations at $q = 1$, each with its mean fitted so the external sky sits at $P(\text{empty}) = 5\%$. Dots at today give $F(t_0)$; dots on the 50% line give the mean emergence, 0.18, 0.92 and 2.9 Gyr from now. The grey horizontal line marks one zone in a 10^{10}-zone galaxy.

These rows assume independent normal durations (Appendix A; Figure 6); tail shape and correlations between lineages change them. Correlation leaves the expected count $\Lambda F(t_0)$ unchanged at a fixed marginal percentile, but changes the probability that no lineage has yet turned loud. Two points. First, we sit far out in the tail: at the 1% spread, among the first $\sim 10^{-11}$ of zones to emerge, with the typical zone emerging 0.9 Gyr from now. Second, one tail position covers home and away. At spreads of 100 Myr and up, external galaxies bind harder than ours — many galaxies, each with 10^{10} draws, the nearer ones dominating μ by count — and a position deep enough for them leaves 0.06 expected loud lineages ahead of us at home. At 30 Myr the home condition binds instead, at $F(t_0) \sim 10^{-10}$. Either way the picture is a first flush: a thin leading edge arriving around now in galaxy after galaxy, the bulk behind it. Eventual emergence can still be the rule; the sky charges for it in percentile rather than in q. The flush also has a near date. On the 1% branch, the same arithmetic gives a conditional median wait of about 20 Myr for the first external front to reach us, and we would see some neighbouring galaxies turn loud before their fronts arrived. An empty sky, on this account, is a brief condition.

$F(t_0)$ also gives the cleanest comparison between the two accounts. Both deliver a tiny emerged share today, and both put our luck somewhere. Under hard steps, F is a success share within planetary lifetimes: almost every zone fails. For a single same-aged cohort its cumulative chance scales as t^n; on a cosmic clock at 13.8 Gyr with $n = 5$, one more Gyr multiplies F by $(14.8/13.8)^5 \approx 1.42$. That is a scale comparison, not a demographic forecast over zones of mixed ages. Under the process clock, the eventual emerged share of suitable zones is close to one and $F(t_0)$ is small only because today sits on the leading edge. At the 1% spread it climbs from 6.2×10^{-12} to one half in 0.92 Gyr, a factor of 8×10^{10}. One lottery decides whether a zone ever gets there within its planet's lifetime; the other decides where a zone sits in a race that almost every suitable zone eventually finishes. Their futures therefore differ. Hard steps keep independent emergences sparse over the next Gyr, even though an approaching grabby

front remains possible. The process clock sends an emergence wave through existing zones over the next few Gyr unless earlier arrivals occupy them first. A rising rate of nearby independent emergences is the distinguishing signature, not contact by itself.

We have not converted any of these percentiles into a probability for our existence. Observers only ever sample emerged lineages in unoccupied territory, and modelling that sampling (Bostrom 2002; Olson and Ord 2021), with the occupation of our own galaxy handled explicitly, belongs to a longer treatment.

6. Conclusions

The process clock itself is the largest thing left out: we assume a characteristic duration and inherited progress, and derive neither from chemistry. Within Section 5, the unidentified left tail of the duration law is the main modelling choice; a physical minimum duration would change the percentiles and arrival times qualitatively. Also assumed: a lineage that turns loud stays recognisable once it has crossed its galaxy; counted trajectories are never destroyed and always find a large-enough next arena; each either runs the full course or contributes nothing.

For the 14 Myr cap only the nearest few megaparsecs enter, and there flat, unclustered galaxies are a fair first pass — though the actual Local Group arrangement would shift the cap by factors of order unity. Section 3 uses the same flat machinery over cosmological distances and should be read as a scaling statement until redone in an expanding universe with an evolving galaxy population. Clustering also correlates neighbours' enrichment histories and hence their start times.

Our galaxy enters Section 5 through the expected count of earlier loud lineages and the probability of none under independent durations. A full treatment needs positions within the galaxy, and the selection effect that independent civilisations only report from unoccupied space. Those ingredients would also turn emergence percentiles into observer probabilities. They are deferred, with any attempt to derive durations from accumulated opportunity.

To summarise the account: planet-bound clocks, with each planet running its own abiogenesis and long evolution from scratch, make the sky empty by making loud lineages vanishingly rare per galaxy. That is the pressure that pushes the explanation toward intrinsic hard steps. A shared cosmic start with inherited progress makes the sky empty a different way. With a single shared emergence date and appreciable loud yields, the permitted lead is about 14 Myr. Spreading the starts permits a longer lead, but still places us in the first flush of arrivals on cosmological timescales. With independent, normally distributed durations and $q = 1$, the examples place us among the first roughly 10^{-10} to 10^{-13} of zones to emerge, with the mean emergence still to come. Correlated histories can change probabilities at a fixed marginal percentile. In the independent normal-duration model with a 1% spread and $q = 1$, the fitted timing leaves an expected 0.06 earlier loud lineages in the Milky Way, giving about a 94% chance of none. On that branch the first flush is local as well as global, and the first external front has a conditional median arrival only about 20 Myr from now. Nothing here shows the process clock is true. It shows the clock can be run without a filter, and states the timing it then requires.

Data and code availability

Configurations, saved outputs and verification files sit in the accompanying repository. `make baseline`, `make inference` and `make start-spread` regenerate the sweep numbers; `uv run python -m lineage_cap.iid_tail` regenerates the normal-tail and first-arrival table at `results/iid_tail/normal_tail_estimate.csv`; `MPLCONFIGDIR=.mplconfig uv run python figures/make_note.py` regenerates the figures. `make check` runs the tests.

This note, its figures and data are licensed under CC BY 4.0; the code is MIT licensed. An interactive version of the calculation is at https://www.fergusfettes.com/lineage-cap/.

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Appendix A. Calculation

Times are cosmic ages in Gyr; distances are Mpc; $c = 306.6014$ Mpc Gyr^{-1}. Effective trajectory counts grow as

$$N(t) = N_0 \,\mathrm{clip}\left(\frac{t-s}{t_0-s}, 0, 1\right)^{\alpha},$$

with $s = 0.2$ Gyr, $\alpha = 1$, $N_0 = 10^{10}$ at the centre of a 10^8–10^{11} sweep. Emerged counts by deadline t under the two clocks:

$$m_{\mathrm{planet}}(t;T) = N(t-T), \qquad m_{\mathrm{inherited}}(t;f) = \mathbf{1}_{t\geq f} N(t),$$

where $f = s + T = t_0 - \delta$. A galaxy with m emerged lineages hosts a loud one with probability

$$p_g(t) = 1 - (1-q)^{m(t)},$$

evaluated as $-\mathrm{expm1}[m \log 1p(-q)]$. Visibility and arrival impose deadlines

$$t_{\mathrm{vis}}(r) = t_0 - D/v - r/c, \qquad t_{\mathrm{arr}}(r) = t_0 - r/v,$$

with $D = 0.03$ Mpc and launch delay set to zero. Writing a deadline as $t_k = t_0 - d_k - r/c_k$,

$$\mu_k = 4\pi n_g \int_0^{R_k} r^2 p_g[t_k(r)]\, dr, \qquad P(\mathrm{empty}) = e^{-\mu_k}.$$

The working line is $\mu_k = -\log(0.05)$. When every galaxy in reach is loud, $p_g = 1$ inside the integral and

$$\delta_{\max} = d_k + \left(\frac{3\mu_k}{4\pi n_g c_k^3}\right)^{1/3},$$

which is the 13.7 Myr plateau: $d_{\mathrm{vis}} = D/v \approx 0.2$ Myr plus the light-travel radius that encloses an average of three galaxies.

For starts spread uniformly over width W with one draw $S = s_{\text{first}} + U$ per galaxy and a common duration, the share of galaxies emerged by t is

$$F(t) = \text{clip}\left(\frac{t - t_0 + \delta}{W}, 0, 1\right),$$

and averaging over the start draw comes before the at-least-one step: $p_g = F[1 - (1-q)^{N(t)}]$. With H independent start groups per galaxy, $p_g = 1 - \{1 - F[1 - (1-q)^{N(t)/H}]\}^H$.

The per-lineage estimates in Section 5 use a single shared start S for all galaxies and durations $D \sim \mathcal{N}(m, \sigma_T^2)$ drawn independently per zone. Writing F_D for the duration CDF, the emerged share by t is $F(t) = F_D(t - S)$ and a galaxy is loud by deadline t with probability

$$p_g(t) = 1 - [1 - qF(t)]^{N(t)}.$$

Inserting this in the visibility integral and solving $\mu_{\text{vis}} = -\log(0.05)$ for m at fixed σ_T gives the emerged shares $F(t_0)$ tabulated in Section 5 ($q = 1$, $N(t_0) = 10^{10}$). The Milky Way column is $qN(t_0)F(t_0)$, the mean count of loud lineages by t_0 in one galaxy; the probability that a named lineage emerging at t_0 is the first loud one in its galaxy is $\exp[-qN(t_0)F(t_0)]$ to Poisson accuracy, with travel-time corrections across the galaxy below 1% at these spreads. For first arrival, the same p_g is integrated with the physical-arrival deadline: a galaxy at r must have emerged by $t_0 + \Delta - r/v$. Solving for the Δ at which the expected arrival count since t_0 reaches $\log 2$ gives the conditional median wait quoted in Section 5 (19.6 Myr at $\sigma_T = 136$ Myr). Quadrature over lookback time with the normal CDF reproduces the table; values are saved in `results/iid_tail/normal_tail_estimate.csv`. Tail shape matters at the order-of-magnitude level and these rows should be read as normal-tail arithmetic.

Sweep integrals use Gauss–Legendre quadrature with a change of variables that resolves the thin partly-loud shell at the edge of the visible volume; inversions use bisection. Doubling the quadrature order shifts saved boundaries by at most 1.4×10^{-7} (500 baseline boundaries) and 1.1×10^{-7} (600 start-window boundaries, 60 probability ceilings) in relative terms. Those figures describe the arithmetic. The scenario choices are uncertain by far more.